Yiming Gao

Ph.D. Student in Cyberspace Security, University of Science and Technology of China
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Research Interests

Lattice algorithms and discrete Gaussian sampling; algorithmic number theory and provable cryptanalysis; subset-sum and related computational problems.

Education

University of Science and Technology of China (USTC) — Hefei, China
Ph.D. in Cyberspace Security; Advisor: Prof. Honggang Hu — *Sep. 2023 – Present*

University of Science and Technology of China (USTC) — Hefei, China
B.Eng. in Information Security, GPA: 3.47/4.30 — *Sep. 2019 – Jun. 2023*

Peer-Reviewed Publications

1. **Better Bounds for Finding Fixed-Degree Isogenies via Coppersmith's Method**
Marius A. Aardal, Diego F. Aranha, Yansong Feng, **Yiming Gao**, and Yanbin Pan.
EUROCRYPT 2026. [DOI] *Authors are listed alphabetically.*

2. **On Factoring and Power Divisor Problems via Rank-3 Lattices and the Second Vector**
Yiming Gao, Yansong Feng, Honggang Hu, and Yanbin Pan.
Mathematics of Computation, 2026. [DOI]

3. **Attacking ECDSA with Nonce Leakage by Lattice Sieving: Bridging the Gap with Fourier Analysis-Based Attacks**
Yiming Gao[†], Jinghui Wang[†], Honggang Hu, and Binang He.
ASIACRYPT 2024. [DOI] [†]*Equal contribution.*

Preprints

1. **Gaussian Difference Dynamics: Breaking the $2^{n/2}$ Barrier for the Shortest Vector Problem and Discrete Gaussian Sampling**
Yiming Gao, Xuyuan Han, and Honggang Hu.
Preprint, 2026. [PDF]

2. **Faster Matrix Multiplication Leads to Faster Lattice Sieving**
Yiming Gao and Honggang Hu.
IACR ePrint, 2026. [ePrint]

3. **ℓ_∞ Norm SVP in $\widetilde{O}(3^n)$**
Yiming Gao.
Preprint, 2026. [PDF]

4. **Discrete Gaussian Sampling Meets BDGL Decoding: Solving the Shortest Vector Problem in $2^{0.5596n+o(n)}$ Time**
Yiming Gao, Yansong Feng, and Honggang Hu.
IACR ePrint, 2026. [ePrint]

5. **Solving the Shortest Vector Problem in $2^{0.7314n+o(n)}$ Time via Discrete Gaussian Sampling on Superlattices**
Yiming Gao, Yansong Feng, and Honggang Hu.
IACR ePrint, 2026. [ePrint]

6. **Faster SVP in Polynomial Space**
Yansong Feng, **Yiming Gao**, and Jiaqi Liu.
IACR ePrint, 2026. [ePrint]

7. **Provable Subexponential Algorithms for NIST Third-Round Lattice Families**
Yiming Gao, Xuyuan Han, and Honggang Hu.
IACR ePrint, 2026. [ePrint]

Preprints (continued)

8. **A Provable Subexponential-Time Algorithm for LWE from $n + o(n)$ Samples via Wagner-Style Gaussian Sampling**
   Xuyuan Han, **Yiming Gao**, and Honggang Hu.
   IACR ePrint, 2026. [ePrint]

9. **Provable Recovery of RSA Private Exponents below $N^{11/42-\varepsilon}$**
   Yiming Gao and Honggang Hu.
   IACR ePrint, 2026. [ePrint]

10. **Subset Balancing and Generalized Subset Sum via Lattices**
    Yiming Gao, Yansong Feng, Honggang Hu, and Yanbin Pan.
    arXiv preprint, 2026. [arXiv]

Selected Talks

Rank-3 Lattices and the Second Vector in Deterministic Factoring Sep. 2026
Invited talk, Academy of Mathematics and Systems Science, Chinese Academy of Sciences (online).

Better Bounds for Finding Fixed-Degree Isogenies via Coppersmith's Method May 2026
Conference talk, EUROCRYPT 2026.

Attacking ECDSA with Nonce Leakage by Lattice Sieving: Bridging the Gap with Fourier Analysis-Based Attacks
Invited talk, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China. Nov. 2025
Conference talk, ASIACRYPT 2024. Dec. 2024

Academic Service

External Reviewer: 67th IEEE Symposium on Foundations of Computer Science (FOCS 2026)

Technical Skills

Programming: C, Python, SageMath **Tools:** Linux, shell scripting, LaTeX